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S. No. of Question Paper : 2816

Unique Paper Code : 32351403

GC-4

Name of the Paper : C10-Ring Theory and Linear

Algebra-I

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any *three* parts from each question.

All questions are compulsory.

1. (a) Let R be a ring. Prove that the center of R is a subring of R . Is Z_6 a subring of Z_{12} ? 3+2=5

(b) Show that a finite integral domain is a field. Give an example of the same. Also, give an example of an infinite integral domain which is not a field. 3+1+1=5

P.T.O.

- (b) Using $v = \ln u$ and $v = f(x) + g(y)$, find the solution of the Cauchy problem

$$y^2 u_x^2 + x^2 u_y^2 = (xyu)^2, \quad \text{with } u(x,0) = e^{x^2}. \quad (7\frac{1}{2})$$

- (c) Find the integral surfaces of the equation $uu_x + u_y =$ for the following initial data: $x(s,0) = s$, $y(s,0) = 2$, $u(s,0) = s$. (7\frac{1}{2})

SECTION - II

2. Attempt any **one** part out of the following :

- (a) State Conservation Law and derive the Burger Equation

$$u_t + uu_x = \nu u_{xx},$$

where ν is the kinematic viscosity and $u(x,t)$ is the fluid velocity field. (6)

- (b) Derive the damped wave equation of a string

$$u_{tt} + au_t = c^2 u_{xx},$$

where the damping force is proportional to the velocity and a is constant. Considering a restoring force proportional to the displacement of a string, show that the resulting equation is

$$u_{tt} + au_t + bu = c^2 u_{xx}, \quad \text{where } b \text{ is a constant.} \quad (6)$$

Attempt any two parts out of the following :

- (a) Find the characteristics, characteristic coordinates and reduce the equation given below to the canonical form :

$$u_{xx} + x^2 u_{yy} = 0, \text{ for } x \neq 0. \quad (6)$$

- (b) Use the polar coordinates r and θ ($x = r \cos \theta$, $y = r \sin \theta$) to transform the Laplace equation $u_{xx} + u_{yy} = 0$ into the polar form

$$\nabla^2 u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0. \quad (6)$$

- (c) Obtain the general solution of the equation given below :

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} + xy u_x + y^2 u_y = 0 \quad (6)$$

SECTION - III

Attempt any three parts out of the following :

- (a) Determine the solution of the Cauchy problem given below :

$$u_{tt} - 9u_{xx} = 0, \quad x \in \mathbb{R}, \quad t > 0$$

$$u(x, 0) = \cos x, \quad u_t(x, 0) = \sin 2x \quad (7)$$

(b) Find the solution of the initial boundary- value problem.

$$u_{tt} = 4u_{xx}, \quad 0 < x < 1, \quad t > 0,$$

$$u(x, 0) = 0, \quad 0 \leq x \leq 1,$$

$$u_t(x, 0) = x(1-x), \quad 0 \leq x \leq 1,$$

$$u(0, t) = 0, \quad u(1, t) = 0, \quad t \geq 0. \quad (7)$$

(c) Solve the Cauchy problem for the non- homogeneous wave equation

$$u_{tt} = c^2 u_{xx} + h^*(x, t),$$

with the initial conditions

$$u(x, 0) = f(x), \quad u_t(x, 0) = g^*(x) \quad (7)$$

(d) Solve the characteristic initial- value problem

$$xu_{xx} - x^3u_{yy} - u_x = 0, \quad x \neq 0,$$

$$u(x, y) = f(y) \quad \text{on} \quad y - \frac{x^2}{2} = 0 \quad \text{for} \quad 0 \leq y \leq 2,$$

$$u(x, y) = g(y) \quad \text{on} \quad y + \frac{x^2}{2} = 4 \quad \text{for} \quad 2 \leq y \leq 4,$$

where $f(2) = g(2)$. (7)

SECTION - IV

Attempt any **three** parts out of the following :

(a) Discuss the solution of the Heat Conduction Problem

$$u_t = ku_{xx}, \quad 0 < x < l, \quad t > 0$$

$$u(0,t) = 0, \quad t \geq 0$$

$$u(l,t) = 0, \quad t \geq 0$$

$$u(x,0) = f(x), \quad 0 \leq x \leq l. \quad (7)$$

(b) Solve using the method of separation of variables

$$u_{tt} - c^2 u_{xx} = 0, \quad 0 < x < \pi, \quad t > 0,$$

$$u(x,0) = \sin x, \quad u_t(x,0) = x^2 - \pi x, \quad 0 \leq x \leq \pi,$$

$$u(0,t) = u(\pi,t) = 0, \quad t > 0. \quad (7)$$

(c) Determine the solution of

$$u_{tt} = c^2 u_{xx} + A \sinh x, \quad 0 < x < l, \quad t > 0,$$

$$u(x,0) = 0, \quad u_t(x,0) = 0, \quad 0 \leq x \leq l,$$

$$u(0,t) = h, \quad u(l,t) = k, \quad t > 0$$

where h , k and A are constants. (7)

- (d) Find the temperature distribution in a rod of length l .
The faces are insulated, and the initial temperature distribution is given by $x(l - x)$.